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Example: 12.15

Differentiate the function Y • $3x^2$ • $16x^3$ with respect to x.

Solution:

$$Y \bullet 3x^2 \bullet 16x^3$$

Differentiating,

$$\frac{dy}{dx} = 3(2)x^{2-1} + 16(3)x^{3-1}$$

$$= 6x^1 + 48x^2$$

$$\frac{dy}{dx} = 6x + 48x^2$$

Example: 12.16

If Y • $2x^3$ • 6x, then find $\frac{dy}{dx}$

Solution:

Y • $2x^3$ • 6x

Differentiate 'y' with respect to x,

$$\frac{dy}{dx} = 2(3)x^{3-1} - 6(1)x^{1-1}$$

$$= 6x^2 - 6x^0$$

$$\frac{dy}{dx} = 6x^2 - 6$$

12.4.3 Application of Differential Calculus

The relation between two or more variables can be expressed by means of a function. Continuous functions alone are differentiable. For instance, the differential calculus is applicable for finding the following:

(1) The rate of change in demand with respect to price (in micro economics)

(2) The rate of change in income with respect to the investment. (in macroeconomics) .

12.4.4 Marginal concepts

Marginal concept is concerned with variations of Y (on the margin of X), that is, it is the variation corresponding in Y to a very small variation in X. (X is the independent variable and Y is the dependent variable)

12.4.5 Marginal Product

Marginal product of a factor of production refers to addition to total product due to the use of an additional unit of a factor.

$$MP = d(TP)/dQ = \Delta TP/\Delta Q$$

12.4.6 Marginal Cost

Marginal cost is an addition to the total cost caused by producing one more unit of output. In symbols:

$$MC = \frac{d(TC)}{dQ} \text{ or } MC = \frac{\Delta TC}{\Delta Q}$$

Where, ΔTC represents a change in total cost and ΔQ represents a small change in output or quantity. (in economics one worker, one output etc are assumed to be very small units)

Example: 12.17

Given the total cost function, TC • 15 • $3Q^2$ • $7Q^3$, drive the marginal cost function.

Solution:

$$TC \bullet 15 \bullet 3Q^2 \bullet 7Q^3$$

$$MC = \frac{d(15)}{dQ} + \frac{d(3Q^2)}{dQ} + \frac{d(7Q^3)}{dQ}$$

$$\bullet\ 0 \bullet 3(2)Q^{2\cdot 1} \bullet 7(3)Q^{3\cdot 1}$$

$$MC \bullet 6Q \bullet 21Q^2$$